15.5

DATA SET

DS1518

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

	Training Period (hours)			
	.5	1.0	1.5	2.0
1	8	9	4	4
2	14	7	6	7
3	9	5	7	5
4	12	8	8	5
5	10	9	6	4

$k = 4.$

Use the Kruskal–Wallis H-Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

① $\begin{cases} H_0: & \text{分佈一致} \\ H_a: & \text{分佈不一致} \end{cases}$

② $\alpha = 0.01$

$n = 20$

③ rank

.5	1.0	1.5	2.0
$\cancel{8}$ 13	$\cancel{9}$ 16	$\cancel{4}$ 2	$\cancel{4}$ 2
14 20	$\cancel{7}$ 10	$\cancel{6}$ 7.5	$\cancel{7}$ 10
$\cancel{9}$ 16	$\cancel{5}$ 5	$\cancel{7}$ 10	$\cancel{5}$ 5
12 19	8 13	$\cancel{8}$ 13	$\cancel{5}$ 5
10 18	$\cancel{9}$ 16	$\cancel{6}$ 7.5	$\cancel{4}$ 2

驗算 $86+60+40+24 = 210$ ✓

$\frac{(1+20)\cdot 20}{2} = 210$ ✓

$T_1 = 86$ $T_2 = 60$ $T_3 = 40$ $T_4 = 24$

④ $H^* = \frac{12}{20 \times 21} \cdot \left(\frac{86^2}{5} + \frac{60^2}{5} + \frac{40^2}{5} + \frac{24^2}{5}\right) - 3 \times (20+1) = 12.269$

⑤ RR: $\chi^2_{0.01,3} = 11.3449$

∴ ⑥ reject H_0 #

15.5

DATA SET DS1520

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10–19	20–39	40–59	60–69
	29 21	24 8	37 39	28 18
	33 29.5	27 15	25 10.5	29 21
	26 12.5	33 29.5	22 5.5	34 34
	27 15	31 24	33 29.5	36 37.5
	39 40	21 3	28 18	21 3
	35 36	28 18	26 12.5	20 1
	33 29.5	24 8	30 23	25 10.5
	29 21	34 34	34 34	24 8
	36 37.5	21 3	27 15	33 29.5
	22 5.5	32 25.5	33 29.5	32 25.5
Total	309	275	295	282
	$T_1=247.5$	$T_2=168$	$T_3=216.5$	$T_4=188$

$k=4$ Treatment

$n=40$

驗算: 810 $= (1+40)\cdot 20 = 810$

a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal–Wallis H-test with $\alpha = .01$.

b. Find the approximate p-value for the test in part a.

c. Since the F-test in Chapter 11 and the H-test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F-test produced the test statistic $F = 0.87$ with p-value $= 0.468$, compare this p-value with the p-value in part b and explain the implications of the comparison.

(a)

① $\begin{cases} H_0: \text{ no significant difference} \\ H_a: \text{ significant difference} \end{cases}$

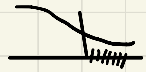
② $\alpha = 0.01$

③ $H_{STAT} = \frac{12}{n(n+1)} \sum \frac{T_i^2}{n_i} - 3\cdot(n+1)$

④ $H^* = \frac{12}{40\times 41} \times \left(\frac{247.5^2 + 168^2 + 216.5^2 + 188^2}{10} \right) - 3\times 41$

$= 2.6316$

⑤ RR: $\chi^2_{0.01,3} = 11.3449$



⑥ ∴ non-reject H_0.

(b) p-value $= P(\chi^2_3 > 2.6316)$ 介在 0.1~0.9 之間

> 0.05

∴ non-reject H_0

(c) $F^* = 0.87$ p-value $= 0.468 > 0.05 \rightarrow$ non-reject H_0

兩者結論相同, 皆不拒絕H_0.

∴ ① data might not severely violate $\begin{cases} \text{常態性} \\ \text{or} \\ \sigma^2 \text{不一致性} \end{cases}$

or

15.6

10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.[7] The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

DATA SET DS1528

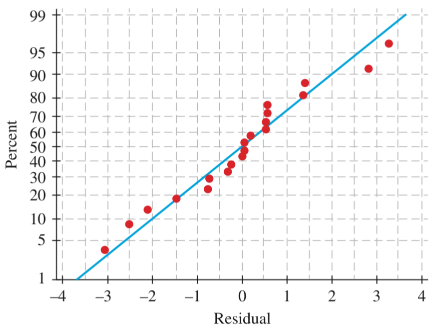
BLOCK — Antibiotic — Treatment

Child	1	2	3	4
1	4.8 2	2.2 1	6.8 4	6.2 3
2	8.1 3	9.2 2	6.6 1	9.6 4
3	5.0 3	2.6 1	3.6 2	6.5 4
4	7.9 2	9.4 4	5.3 1	8.5 3
5	3.9 3	7.4 4	2.1 2	2.0 1
	13	13	10	15

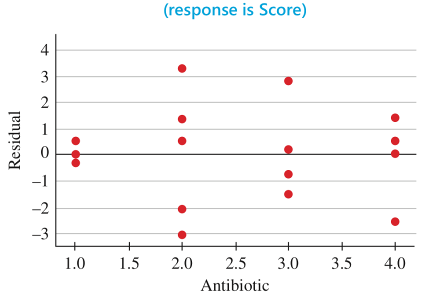
a. What design is used in collecting these data?

b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?

Normal Probability Plot
(response is Score)



Residuals Versus Antibiotic
(response is Score)



c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

(a.) randomised block design

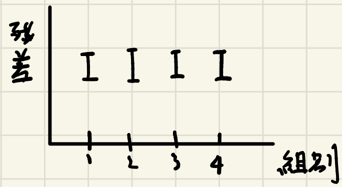
Blocks: 5

Treatment: 4

(b.) ① Normal Probability Plot: not normal

② σ^2一致性的話, residual 分散程度應要相同 $\boxed{\text{上下寬度相同}}$

應要張



(c). Fr 檢定

① $\begin{cases} H_0 : \text{分佈一致} \\ H_a : \text{有分佈不一致} \end{cases}$

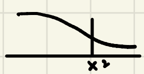
② $\alpha = 0.05$

③ $Fr_{STAT} = \frac{12}{b \cdot k(k+1)} \sum (T_i)^2 - 3 \times b \times (k+1)$

④ $Fr^* = \frac{12}{5 \times 4 \times 5} \sum (12^2 + 13^2 + 10^2 + 15^2) - 3 \times 5 \times 5$

$= 1.56$

⑤ RR:



$\chi^2_{3, 0.05} = 7.8143$

⑥ ∴ non-reject H_0.

15.6

DATA SET DS1529

11. Yield of Wheat Exercise 9 (Chapter 11 Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

$k=5$ $b=6$

Variety	Location 1	2	3	4	5	6	
A	35.3 4	31.0 2	32.7 3	36.8 3	37.2 3	33.1 3	18
B	30.7 1	32.2 3	31.4 2	31.7 2	35.0 2	32.7 2	12
C	38.2 5	33.4 4	33.6 4	37.1 4	37.3 4	38.2 5	26
D	34.9 3	36.1 5	35.2 5	38.3 5	40.2 5	36.0 4	27
E	32.4 2	28.9 1	29.2 1	30.7 1	33.9 1	32.1 1	7

a. Use the Friedman F_r-test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.

b. When the analysis of variance was performed in Chapter 11, the ANOVA F-test produced the test statistic $F = 18.61$ with p-value $= .000$. How do these results compare with results of the Friedman F_r-test in part a? Explain.

a.

① $\begin{cases} H_0: \text{no difference} \\ H_a: \text{difference} \end{cases}$

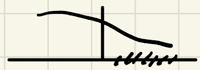
② $\alpha = 0.05$

③ $F_{r\,STAT} = \dfrac{12}{b\cdot k(k+1)} \sum (T_i)^2 - 3\cdot b\cdot(k+1)$

④ $F_r^* = \dfrac{12}{6\cdot 5\cdot 6} \cdot (18^2+12^2+26^2+27^2+7^2) - 3\cdot 6\cdot 6$

$= 20.133\sim$

⑤ RR:

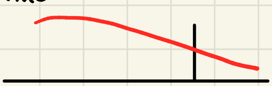


$\chi^2_{4,\,0.05} = 9.48773$

⑥ ∴ reject H_0.

b. p-value = 0

RR:



$\boxed{\text{F: 永遠右尾}}$

$F_{(5-1),(5-1)(6-1),0.05} = F_{4,20,0.05} = 5.86$ ∴ reject H_0.

→ 不論是否 normal 皆會 reject H_0

15.7

DATA SET

DS1530

11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

$n=12.$

Voter	Rating	x	Distance	y	$\lvert d \rvert$
1	12	7.5	75	3	4.5
2	7	4	165	7	3
3	5	3	300	12	9
4	19	12	15	1	11
5	17	10	180	8	2
6	12	7.5	240	11	3.5
7	9	6	120	4	2
8	18	11	60	2	9
9	3	1	230	10	9
10	8	5	200	9	4
11	15	9	130	5.5	3.5
12	4	2	130	5.5	3.5

a. Calculate Spearman's rank correlation coefficient r_s.

b. Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

An Abbreviated Version of Table 9 in Appendix I; for Spearman's Rank Correlation Test 右尾

n	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
5	.900	—	—	—
6	.829	.886	.943	—
7	.714	.786	.893	—
8	.643	.738	.833	.881
9	.600	.683	.783	.833
10	.564	.648	.745	.794
11	.523	.623	.736	.818
12	.497	.591	.703	.780
13	.475	.566	.673	.745
14	.457	.545		
15	.441	.525		
16	.425			
17	.412			
18	.399			
19	.388			
20	.377			

$n=12$

$\alpha=0.05$

法一

a. $r_s = 1 - \frac{6}{12\times(12^2-1)} \cdot \sum (d_i)^2$

$= 1 - \frac{6}{12\times 143} \times (4.5^2+3^2+9^2+\dots+3.5^2)$

$= -0.587$

法二

$r_s = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$

用 rank

$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y}$

$S_{xx} = \sum (x_i)^2 - n\bar{x}^2$

$S_{yy} = \sum (y_i)^2 - n\bar{y}^2$

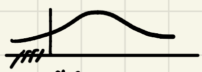
b.

① $\begin{cases} H_0: \rho = 0 \\ H_a: \rho < 0. \end{cases}$

② $\alpha = 0.05$

③ $r_s = -0.587.$

④. RR:



-0.497

⑤. reject H_0.

15.7

DATA SET DS1531

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:[8]

Runner	Years of Competitive Running		10-Kilometer Finish Time (minutes)		$\lvert d_i \rvert$
1	9	7	33.15	1	6
2	13	9	33.33	2	7
3	5	3.5	33.50	3	0.5
4	7	6	33.55	4	2
5	12	8	33.73	5	3
6	6	5	33.86	6	1
7	4	2	33.90	7	5
8	5	3.5	34.15	8	4.5
9	3	1	34.90	9	8

a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.

b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

An Abbreviated Version of Table 9 in Appendix I; for Spearman's Rank Correlation Test

n	$\alpha = .05$	$\alpha = .025$	$\alpha = .01$	$\alpha = .005$
5	.900	—	—	—
6	.829	.886	.943	—
7	.714	.786	.893	—
8	.643	.738	.833	.881
9	.600	.683	.783	.833
10	.564	.648	.745	.794
11	.523	.623	.736	.818
12	.497	.591	.703	.780
13	.475	.566	.673	.745
14	.457	.545		
15	.441	.525		
16	.425			
17	.412			
18	.399			
19	.388			
20	.377			

$n = 9$

a. $r_s = 1 - \frac{6}{n(n^2-1)} \sum (d_i)^2 = 1 - \frac{6}{9 \times 80}(6^2+7^2+0.5^2+2^2+3^2+1+5^2+4.5^2+8^2)$

$= -0.7375$

b.

① $\begin{cases} H_0: \rho = 0 \\ H_a: \rho \neq 0 \end{cases}$

② $\alpha = 0.05$

③ $r_s = -0.7375$

④ RR:

$n = 9$

$\alpha = 0.05/2 = 0.025$.

⑤. ∴ reject H_0